Ahmed Hamda

Engineering Student in Computer Science | Artificial Intelligence | Machine Learning | Software Engineering

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Linkedin Github Portfolio

PROFILE

Third-year engineering student in Web and Multimedia Technologies with a Bachelor's degree in Computer Science specializing in Big Data and Data Analytics. Interested in artificial intelligence, machine learning, deep learning, computer vision, physiological signal processing, data science, and intelligent software systems. Experienced in developing and evaluating machine learning models, preprocessing structured and signal data, and deploying AI models through web and mobile applications.

EDUCATION

Engineering Cycle – Web and Multimedia Technologies **2024 – Present**
Higher Institute of Computer Science and Multimedia of Sfax (ISIMS) *Sfax, Tunisia*

Bachelor's Degree in Computer Science **2021 – 2024**
Big Data and Data Analytics – ISIMS *Sfax, Tunisia*

ACADEMIC AND RESEARCH EXPERIENCE

Bachelor Final-Year Project – ISIMS **2024**

ECG-Based Biometric Identification System

- Developed a biometric identification system based on electrocardiogram (ECG) signals using machine learning and deep learning.
- Preprocessed ECG signals using band-pass filtering, R-peak detection, normalization, and segmentation with R-centered and R-R approaches.
- Extracted signal features including cepstral coefficients, entropy, and Zero Crossing Rate (ZCR).
- Implemented and evaluated Support Vector Machine (SVM) models with linear, polynomial, and RBF kernels.
- Designed and trained a 1D Convolutional Neural Network (1D-CNN) for ECG classification using TensorFlow/Keras.
- Conducted experimental comparison between machine learning and deep learning approaches using the ECG-ID dataset.
- Achieved a maximum identification accuracy of **95.37%** using a CNN with R-centered ECG segmentation.
- Developed a REST API using Flask and integrated the trained model with a React-based visualization and prediction interface.

Keywords: Python, Machine Learning, Deep Learning, ECG, Physiological Signals, Signal Processing, 1D-CNN, CNN, SVM, TensorFlow, Keras, Data Preprocessing, Feature Extraction, Model Training, Model Evaluation, Flask, React, REST API

PROFESSIONAL EXPERIENCE

Summer Internship – NovaSkills **Jul. – Aug. 2026**

ConsultDiag – AI-Powered Full-Stack Business Diagnostic and Consulting Platform

- Designed and developed **ConsultDiag**, a full-stack platform for business diagnostics and consulting services, supporting clients, consultants, and administrators.
- Built a scalable **Angular and FastAPI (Python)** architecture, implementing RESTful APIs and structured data validation using **Pydantic**.
- Developed a versioned questionnaire engine supporting dynamic forms, response collection, progress tracking, and **automated scoring**.
- Integrated **Google Gemini (Generative AI)** to generate and store AI-powered analyses based on structured questionnaire responses.
- Implemented business workflows, secure **JWT authentication and role-based access control (RBAC)**, PostgreSQL data persistence with SQLAlchemy, and file storage using MinIO.

Keywords: Python, FastAPI, Pydantic, Angular, Gemini, Generative AI, LLMs, PostgreSQL, SQLAlchemy, JWT, RBAC, MinIO, Docker

Summer Internship – MEGA PIXEL **Jul. – Aug. 2025**

Full-Stack AI Educational Platform

- Designed and developed a full-stack educational web platform using Flask and Angular.
- Implemented document upload and processing workflows for PDFs and images.
- Integrated AI/NLP-based functionalities to generate summaries, quizzes, and flashcards from uploaded educational content.
- Developed REST APIs and implemented JWT-based authentication.
- Integrated PostgreSQL for persistent data management and Supabase for file storage.
- Built a modular architecture combining backend services, database management, AI processing, and a modern web interface.

Keywords: Python, Flask, Angular, JavaScript, TypeScript, NLP, Artificial Intelligence, REST API, PostgreSQL, Supabase, JWT, Database Management, Full-Stack Development

SELECTED ACADEMIC PROJECTS

VisionTouch – AI-Based Real-Time Object Detection Mobile Application **2025**

- Contributed to the development of a real-time computer vision application using YOLOv8, Flask, and Flutter.
- Built and curated a custom image dataset containing **28 object classes**.
- Fine-tuned a YOLOv8 deep learning model for object detection.
- Deployed the trained model through a Flask REST API for real-time inference on mobile devices.

Keywords: Computer Vision, Object Detection, YOLOv8, Deep Learning, Model Fine-Tuning, Dataset Preparation, Python, Flask, REST API, Real-Time Inference

Customer Churn Prediction Using Machine Learning **2025**

- Developed a supervised machine learning pipeline to predict customer churn from user behavior and subscription data.
- Performed data cleaning, preprocessing, exploratory analysis, and feature engineering.
- Trained and compared machine learning models including Logistic Regression, Random Forest, and XGBoost.
- Evaluated predictive model performance using structured experimental workflows.

Keywords: Machine Learning, Data Science, Data Preprocessing, Feature Engineering, Logistic Regression, Random Forest, XGBoost, Scikit-learn, Pandas, Python

TECHNICAL SKILLS

Programming	Python, Java, JavaScript, TypeScript, C.
Artificial Intelligence	Machine Learning, Deep Learning, Convolutional Neural Networks (CNN), 1D-CNN, Computer Vision, Object Detection, NLP.
Data/Signal Processing	Data Preprocessing, Feature Engineering, ECG Signal Processing, Signal Segmentation, Feature Extraction, Model Evaluation.
ML / Data Libraries	TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib.
Web Development	Flask, Angular, Spring Boot, REST APIs.
Databases	PostgreSQL, MongoDB, MySQL.
Tools	Git, GitHub, Swagger, Postman, Google Colab, VS Code.

AREAS OF EXPERTISE

Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Science, Signal Processing, Software Engineering, Intelligent Systems.

ACADEMIC ACTIVITIES

- **AI Through Python** – Python Club, 2021.
- Participation in technology and innovation events including **Pragma**, **DigiWorld**, and **The AREA**.

LANGUAGES

Arabic	**French**	**English**
Native	Professional proficiency	Professional working proficiency